

Symmetric Difference of Two Sets

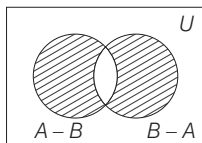
Let A and B be two sets. The symmetric difference of sets A and B is the set $(A - B) \cup (B - A)$ and is denoted by $A \Delta B$ i.e. it is a set consisting of all those members of A which are not in B or all those which are in B but not in A .

Thus, $A \Delta B = \{x : x \in A \text{ but } x \notin B\} \cup \{x : x \in B \text{ but } x \notin A\}$

$$A \Delta B = (A - B) \cup (B - A)$$

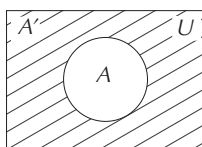
The shaded part in given Venn diagram represents $A \Delta B$.

e.g. If $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and $B = \{1, 3, 5, 6, 7, 8, 9\}$,
 $\therefore A - B = \{2, 4\}$ and $B - A = \{9\}$, then $A \Delta B = \{2, 4, 9\}$.



Complement of a Set

Let U be the universal set and A be any set such that $A \subset U$. Then, the complement of A with respect to U is the set of all those elements of U which are not in A . It is denoted by A' or A^C or $U - A$.



Thus, $A' = \{x : x \in U \text{ and } x \notin A\}$

The shaded portion in the above Venn diagram shows the complement of set A . If A is a subset of the universal set U , then its complement A' is also a subset of U . e.g.

- Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, \dots\}$ and $A = \{1, 3, 5, 7, \dots\}$.

Then, the complement of A is

$$A' = U - A = \{2, 4, 6, 8, \dots\}$$

- If $U = \{a, b, c, d, e, f\}$ and $A = \{a, e\}$,
 then $A' = \{b, c, d, f\}$.

EXAMPLE 4. If $A = \{0, 1, 2, 3, 4\}$, $B = \{1, 2, 3\}$,
 $C = \{3, 5, 7, 9\}$, then $(A - B) \cap (B - C)$ is

- a. $\{1, 2\}$ b. $\{0, 4\}$ c. $\{0, 1, 2, 4\}$ d. ϕ

Sol. d. Given, $A = \{0, 1, 2, 3, 4\}$, $B = \{1, 2, 3\}$ and $C = \{3, 5, 7, 9\}$

$$\therefore A - B = \{0, 4\} \text{ and } B - C = \{1, 2\}$$

$$\therefore (A - B) \cap (B - C) = \phi$$

Properties of Operations on Sets

- $(A')' = A$
- $A \cup A' = U$
- $A \cap A' = \phi$
- $\phi' = U$
- $U' = \phi$
- $A - B = A \cap B'$
- $B - A = B \cap A'$
- $A - B \subseteq A$
- $A - B \neq B - A$
- $A - \phi = A$ and $A - A = \phi$
- $A \subseteq B \Rightarrow A - B = \phi$
- $A - B = B - A \Leftrightarrow A = B$

Note The sets $A - B$, $A \cap B$ and $B - A$ are mutually disjoint sets i.e. the intersection of any of these two sets is an empty set.

Laws of Algebra of Sets

- Idempotent laws** For any set A , we have
 - $A \cup A = A$
 - $A \cap A = A$
- Identity Laws** For any set A ,
 - $A \cup \phi = A$
 - $A \cap U = A$
 i.e. ϕ and U are identity elements for union and intersection, respectively.
- Commutative laws** For any two sets A and B , we have
 - $A \cup B = B \cup A$
 - $A \cap B = B \cap A$
- Associative laws** If A , B and C are any three sets, then
 - $(A \cup B) \cup C = A \cup (B \cup C)$
 - $A \cap (B \cap C) = (A \cap B) \cap C$
- Distributive laws** If A , B and C are any three sets, then
 - $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
 - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- De-Morgan's laws** If A and B are any two sets, then
 - $(A \cup B)' = A' \cap B'$
 - $(A \cap B)' = A' \cup B'$

More Results on Operations on Sets

- If A , B and C are any three sets, then
 - $A - (B \cap C) = (A - B) \cup (A - C)$
 - $A - (B \cup C) = (A - B) \cap (A - C)$
 - $A \cap (B - C) = (A \cap B) - (A \cap C)$
 - $A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C)$
 - $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$
- If A , B and C are finite sets and U be the finite universal set, then
 - $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
 - $n(A \cup B) = n(A) + n(B) \Leftrightarrow A, B$ are disjoint sets.
 - $n(A - B) = n(A) - n(A \cap B)$
 i.e. $n(A - B) + n(A \cap B) = n(A)$
 - Number of elements which belong to exactly one of A or B
 $= n(A \Delta B) = n(A) + n(B) - 2n(A \cap B)$
 - If A , B and C are finite sets, then
 $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$
 - $n(A' \cup B') = n((A \cap B)') = n(U) - n(A \cap B)$
 - $n(A' \cap B') = n((A \cup B)') = n(U) - n(A \cup B)$

EXAMPLE 5. If a set A contains 60 elements and another set B contains 70 elements and there are 50 elements in common, then how many elements does $A \cup B$ contain?

- a. 130 b. 100 c. 80 d. 70

Sol. c. Here, $n(A) = 60$, $n(B) = 70$ and $n(A \cap B) = 50$

$$\begin{aligned} \text{Now, } n(A \cup B) &= n(A) + n(B) - n(A \cap B) \\ &= 60 + 70 - 50 = 130 - 50 = 80 \end{aligned}$$

EXAMPLE 6. A group of 40 people who speaks either english or Hindi, out of these 12 speak both English and Hindi and 22 speak Hindi. How many people speak only english not Hindi?

- a. 30 b. 10 c. 18 d. 28

Sol. c. Let E be the set of students who speak English and H be the set of students who speak Hindi.

$$\text{Given, } n(E \cup H) = 40, \quad n(H) = 22 \text{ and } n(E \cap H) = 12$$

$$\begin{aligned} \therefore n(E \cup H) &= n(E) + n(H) - n(E \cap H) \\ 40 &= n(E) + 22 - 12 \Rightarrow n(E) = 30 \end{aligned}$$

$$\begin{aligned} \text{Number of people who speak English only} \\ &= n(E) - n(E \cap H) = 30 - 12 = 18 \end{aligned}$$

Ordered Pairs

An ordered pair consists of two objects or elements in a given fixed order.

If A and B are two sets and $a \in A$, $b \in B$, then the ordered pair of elements a and b is denoted by (a, b) .

The natural numbers and their squares can be represented by ordered pair in the following way.

$$(1, 1), (2, 4), (3, 9), (4, 16), \dots$$

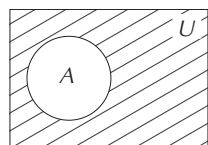
Two ordered pairs (a, b) and (c, d) will be equal, if and only if $a = c$ and $b = d$.

Note $\{a, b\} = \{b, a\}$ but $(a, b) \neq (b, a)$.

PRACTICE EXERCISE

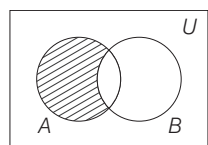
- If $A = \{5, 6, 7\}$ and $B = \{7, 8, 9\}$, then $A \cup B$ is equal to
(a) $\{5, 6, 7, 8, 9\}$ (b) $\{7, 8, 9\}$ (c) $\{5, 6, 7\}$ (d) ϕ
- Given that $A = \{2, 6, 8, 9\}$, $B = \{7, 8, 9, 12\}$, then $B - A$ is equal to
(a) $\{7, 8, 9, 12\}$ (b) $\{7, 12\}$
(c) $\{2, 6, 7, 8, 9, 12\}$ (d) $\{2, 6, 8, 9, 12\}$
- If U is the universal set of all natural numbers and $A = \{1, 2, 3, 4, 5\}$, then compute $A \cap U$.
(a) $\{1, 2, 3, 4\}$ (b) ϕ (c) $\{1, 2, 3, 4, 5\}$ (d) U
- The set $\{2, 4, 16, 256, \dots\}$ can be represented as which one of the following?
(a) $\{x \in N \mid x = 2^{2^n}, n \in N\}$
(b) $\{x \in N \mid x = 2^{2^n}, n = 0, 1, 2, \dots\}$
(c) $\{x \in N \mid x = 2^{4^n}, n = 0, 1, 2, \dots\}$
(d) $\{x \in N \mid x = 2^{2^n}, n = 0, 1, 2, \dots\}$
- If P and Q are any two sets and $P \subset Q$, then
(a) $P \cap Q = \phi$ (b) $P' \cap Q = P$ (c) $P \cap Q = P$ (d) $P \cap Q = Q$
- Which one of the following is a true statement?
(a) $(A - B) \cap (B - A) = \phi$ (b) $(A - B) \cap (B - A) = A$
(c) $(A - B) \cap (B - A) = U$ (d) $(A - B) \cap (B - A) = B$
- If P and Q are any two sets, then $P \cup Q = P \cap Q$, if
(a) P is the empty set (b) Q is the empty set
(c) Both P and Q are empty sets
(d) P and Q are non-empty sets
- The number of non-empty proper subsets of set $A = \{2, 5, 7, 10\}$ is
(a) 16 (b) 15 (c) 14 (d) 8
- Which of the following statements is false for the sets A, B and C , where
 $A = \{x \mid x \text{ is letter of the word 'BOWL'}\}$
 $B = \{x \mid x \text{ is a letter of the word 'ELBOW'}\}$
 $C = \{x \mid x \text{ is a letter of the word 'BELLOW'}\}$
(a) $A \subset B$ (b) $B \supset C$
(c) $B \neq C$ (d) B is a proper subset of C
- If A and B are two sets, then $A \cap (A \cup B)$ equals
(a) A (b) B
(c) ϕ (d) None of these
- The smallest set B such that
 $B \cup \{1, 2\} = \{1, 2, 3, 5, 9\}$ is
(a) $\{3, 5, 9\}$ (b) $\{3, 5, 8\}$
(c) $\{1, 2, 3\}$ (d) None of these
- If $U = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $P = \{0, 1, 2, 3\}$,
 $Q = \{2, 3, 4, 5\}$, and $R = \{4, 5, 6\}$, then $Q' \cap (P \cup R)$ is equal to
(a) $\{0, 1, 6\}$ (b) $\{0, 1, 2, 3, 4, 5\}$
(c) $\{6, 7, 8, 9\}$ (d) $\{2, 3\}$
- If P is a non-empty set, then $(P')'$ is equal to
(a) ϕ (b) U (c) $U - P$ (d) P

14. The shaded region in the adjoining diagram is



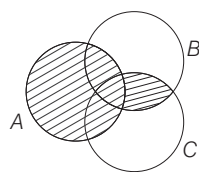
- (a) $A \cup A'$ (b) U (c) A' (d) $A \cap A'$

15. The shaded region in the adjoining diagram represents



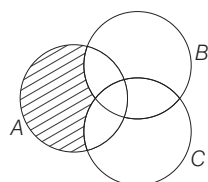
- (a) $(A \cap B)'$ (b) $A \cup B$ (c) $A - B$ (d) $A \cap B$

16. The shaded portion in the following Venn diagram represents



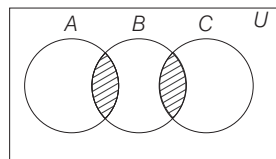
- (a) $A \cup (B \cap C)$ (b) $A \cup (B \cup C)$
(c) $(A \cap B) \cap C$ (d) $(A \cap B) \cup C$

17. The shaded region in the adjoining diagram is



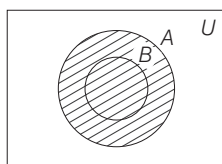
- (a) $A \cap (B - C)$ (b) $A - (B \cup C)$
(c) $A \cap (B \cup C)$ (d) $A \cup (B \cap C)$

18. In the Venn diagram below, shaded portion represents

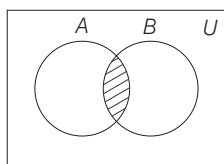


- (a) $A \cup B \cup C$ (b) $A \cap B \cap C$
(c) $(A \cap B) \cup (B \cap C)$ (d) $A \cup B \cap C$

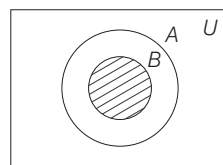
19. The Venn diagram for $A \cup B$ when $B \subset A$ is



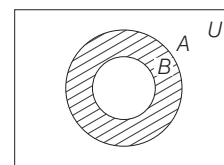
(a)



(b)



(c)



(d)

20. Which one of the following is a correct statement?

- (a) $\phi \in \phi$ (b) $\phi \notin P(\phi)$ (c) $\phi = P(\phi)$ (d) $\phi \in P(\phi)$

21. Let $A = \left\{ 3, \pi, \sqrt{2}, \frac{2}{7}, -5, 3 + \sqrt{7} \right\}$. The subset of A containing all the elements from it which are irrational numbers is

- (a) $\left\{ 3, \frac{2}{7}, -5 \right\}$ (b) $\left\{ 3, \pi, \frac{2}{7}, -5, 3 + \sqrt{7} \right\}$
(c) $\{ \pi, \sqrt{2}, 3 + \sqrt{7} \}$ (d) $\{ 3, -5 \}$

22. If $U = \{x : x \in N\}$, $A = \{x : x \text{ is an odd number}\}$, then A' is equal to

- (a) $\{x : x \text{ is an even number}\}$ (b) $\{x : x \text{ is an odd number}\}$
(c) $\{x : x \text{ is a natural number}\}$ (d) $\{x : x \text{ is an integer}\}$

23. If two sets are disjoint, then their intersection is

- (a) null set (b) singleton
(c) a infinite set (d) None of these

24. If A and B are two non-empty sets, then $A - (A - B)$ equals

- (a) B (b) $A - B$ (c) $A \cap B$ (d) $A' \cap B'$

25. Let two sets A and B have $2n$ and $4n$ elements respectively, where n is a natural number. What can be the minimum number of elements in $A \cup B$?

- (a) $2n$ (b) $3n$ (c) $4n$ (d) $6n$

26. If $P = \{x : x^2 - 3x + 2 = 0\}$, $Q = \{x : x^2 + 4x - 12 = 0\}$, then $P - Q$ is

- (a) $\{1, 2\}$ (b) $\{2\}$ (c) $\{1\}$ (d) $\{4, 3\}$

27. If $A = \{(2^{2n} - 3n - 1) | n \in N\}$ and

$B = \{9(n - 1) | n \in N\}$, then which one of the following is correct?

- (a) $A \subset B$ (b) $A \subset A$ (c) $A = B$
(d) Neither A is a subset of B nor B is a subset of A

28. If A, B and C are any three sets, then

- (a) $A - (B \cup C) = (A - B) \cup (A - C)$
(b) $A - (B \cup C) = (A - B) - (A - C)$
(c) $A - (B \cup C) = (A - B) \cap (A - C)$
(d) $A - (B \cup C) = (A \cup B) - (A \cup C)$

29. If P and Q are two sets such that

$n(P) = m$, $n(Q) = n$ and $n(P \cap Q) = p$, then $n(P \cup Q)$ is equal to

- (a) $m + n$ (b) $m + n + p$ (c) $m + n - p$ (d) $m - n - p$

30. A and B are two sets such that $n(A) = 17$, $n(B) = 23$, $n(A \cup B) = 38$. Then, $n(A \cap B)$ is

- (a) 40 (b) 78 (c) 2 (d) None of these

- 31.** Let $A = \{x : x^2 - 6x + 8 = 0\}$ and $B = \{x : 2x^2 + 3x - 2 = 0\}$. Then, which one of the following is correct?
 (a) $A \subseteq B$ (b) $B \subseteq A$
 (c) Neither $A \subseteq B$ nor $B \subseteq A$ (d) $A = B$
- 32.** If $A = \{1, 2, 3, 4\}$, $B = \{2, 3, 4, 5\}$, $C = \{1, 3, 4, 5, 6, 7\}$, then $A \cap (B \cup C)$ is equal to
 (a) $\{1, 2, 3, 4, 5, 6, 7\}$ (b) $\{1, 2, 3, 4\}$
 (c) $\{1, 2, 3, 4, 5\}$ (d) $\emptyset$
- 33.** Let the set A and B be given by $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6, 8, 10\}$ and the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, then $(A \cup B)'$ is
 (a) $\{2, 4\}$ (b) U
 (c) $\{1, 3, 5, 6, 7, 8, 9, 10\}$ (d) $\{5, 7, 9\}$
- 34.** What is $\{(A \cup B)' \cap A\} - (A - B)$ equal to?
 (a) $\emptyset$ (b) A (c) B (d) B'
- 35.** Which one of the following is a correct statement?
 I. $\{a\} \in \{\{a\}, \{b\}, \{c\}\}$ II. $\{a\} \subseteq \{\{a\}, b, c\}$
 III. $\{a, b\} \subseteq \{\{a\}, b, c\}$ IV. $a \subseteq \{\{a\}, b, c\}$
 (a) Only I (b) Only II (c) III and IV (d) Only IV
- 36.** Which one of the following is an infinite set?
 (a) $\{x : x \text{ is a whole number less than or equal to } 1000\}$
 (b) $\{x : x \text{ is a natural number less than } 1000\}$
 (c) $\{x : x \text{ is a positive integer less than or equal to } 1000\}$
 (d) $\{x : x \text{ is an integer and less than } 1000\}$
- 37.** Which one of the following is not correct in respect of the sets A and B ?
 (a) If $A \subseteq B$, then $B \cup A = B$
 (b) If $A \subseteq B$, then $A \cap (A - B) = \emptyset$
 (c) If $A \subseteq B$, then $B \cap A = A$
 (d) If $A \cap B = \emptyset$, then either $A = \emptyset$ or $B = \emptyset$
- 38.** If $R \in \overleftrightarrow{PQ}, S \notin \overleftrightarrow{PQ}$ and S does not lie on $\overleftrightarrow{PQ}$ extended, then
 (a) $\overleftrightarrow{PQ} \cap \overleftrightarrow{RS} = \emptyset$ (b) $\overleftrightarrow{PQ} \cap \overleftrightarrow{RS} = \{R\}$
 (c) $\overleftrightarrow{PQ} \cap \overleftrightarrow{RS} = \{S\}$ (d) $\overleftrightarrow{PQ} \cap \overleftrightarrow{RS} = \{P\}$
- 39.** In a group of 1000 people, there are 750 people who can speak Hindi and 400 who can speak English. How many can speak Hindi only?
 (a) 600 (b) 150 (c) 300 (d) 500
- 40.** In a committee 50 people speak French, 20 speak Spanish and 10 speak both Spanish and French. How many speak atleast one of these two languages?
 (a) 60 (b) 50 (c) 30 (d) 70
- 41.** Every student in a class of 42 students, studies atleast one of the subjects, Mathematics, English and Commerce, 14 students study Mathematics, 20 Commerce and 24 English. 3 students study Mathematics and Commerce, 2 English and Commerce and there is no student who studies all the three subjects. The number of students who study Mathematics but not Commerce is
 (a) 4 (b) 3 (c) 12 (d) 11
- 42.** In an examination, 52% candidates failed in English and 42% failed in Mathematics. If 17% candidates failed in both English and Mathematics, what percentage of candidates passed in both the subjects?
 (a) 18% (b) 21% (c) 23% (d) 25%
- 43.** State which of the following statements about sets is/are true?
 I. Every subset of a finite set is finite.
 II. $\emptyset$ is a subset of $\{0\}$.
 Select the correct answer using the codes given below
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 44.** Which of the following is/are examples of empty set?
 I. $A = \{x : x + 3 = 3, x \in I\}$
 II. $B = \{x : x \text{ is a positive even integer and prime}\}$
 III. $C = \{x : x^2 = 16, x \text{ is odd integer}\}$
 Select the correct answer using the codes given below
 (a) Only I (b) I and III (c) Only III (d) I, II and III
- 45.** Consider the following statements :
 I. $A' \cup B = (A \cap B)'$ II. $(\phi)'\cup\phi$
 III. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
 Which of the statement(s) given above is/are correct?
 (a) Only II (b) I and III (c) Only III (d) II and III
- 46.** Let $P =$ Set of all integral multiples of 3
 $Q =$ Set of all integral multiples of 4
 $R =$ Set of all integral multiples of 6
 Consider the following relations
 I. $P \cup Q = R$ II. $P \subset R$ III. $R \subset (P \cup Q)$
 Which of the relation(s) given above is/are correct?
 (a) Only I (b) Only II (c) Only III (d) II and III
- 47.** Consider the following statements:
 I. Set of all points of a given line is a infinite set.
 II. The set of all birds in a zoo is a infinite set.
 III. Good books in a school library is a set.
 Which of the statement(s) given above is/are correct?
 (a) Only I (b) Only II (c) Only III (d) All of these
- 48.** Given that the set $A = \{0, 1, 2, 3\}$, which of the following statements about A are true?
 I. A is a finite set.
 II. A is a subset of the set of integers.
 III. $\{1, 2\}$ is a proper subset of A .
 IV. A is the null set.
 Select the correct answer using the codes given below
 (a) I, II and III (b) I and IV
 (c) I and III (d) All of these

- 49.** State which of the sets given below are infinite set?
 I. Set of all concentric circles.
 II. $\{x : x \text{ is a multiple of } 2, x \text{ is an integer}\}$
 III. The set of lines which are parallel to X -axis.
 IV. The set of positive integers greater than 100.

Select the correct answer using the codes given below

- (a) I and II (b) II and III
 (c) Only I (d) All of these
- 50.** Which of the following sets are equivalent?
 I. $A = \{1, 2, 3, 4, 5\}, B = \{7, 8, 9, 10, 11\}$
 II. $A = \{x, y, z\}, B = \{p, q\}$
 III. $P = \{2, 4, 6, 8\}, R = \{a, b, c, d\}$
 IV. $A = \{36, 39, 42, 45\}, B = \{42, 39, 45, 36\}$

Select the correct answer using the codes given below

- (a) I, II and IV (b) II, III and IV
 (c) I, III and IV (d) None of these

Directions (Q. Nos. 51-52) Answer the questions based on the following information.

In a survey of 250 students, it was found that 150 play cricket, 100 play basketball and 120 play football, further, 30 of them play both basketball and football, 50 play both cricket and basketball and 60 play both cricket and football.

- 51.** The maximum number of students who play all the three sports
 (a) 20 (b) 10 (c) 15 (d) None of these
- 52.** If 5 students play none of the three sports then numbers of students who play at least two sports
 (a) 100 (b) 110 (c) 120 (d) 130

PREVIOUS YEARS' QUESTIONS

- 53.** If $A = \{x : x \text{ is an even natural number}\}$, $B = \{x : x \text{ is a natural number and multiple of } 5\}$ and $C = \{x : x \text{ is a natural number and multiple of } 10\}$, then what is the value of $A \cap (B \cup C)$? **2012 I**
 (a) $\{10, 20, 30, \dots\}$ (b) $\{5, 10, 15, 20, \dots\}$
 (c) $\{2, 4, 6, \dots\}$ (d) $\{20, 40, 60, \dots\}$
- 54.** Which one of the following is a null set? **2012 II**
 (a) $A = \{x \text{ is a real number} : x > 1 \text{ and } x < 1\}$
 (b) $B = \{x : x + 3 = 3\}$
 (c) $C = \{\emptyset\}$
 (d) $D = \{x \text{ is a real number} : x \geq 1 \text{ and } x \leq 1\}$
- 55.** Let $x \in \{2, 3, 4\}$ and $y \in \{4, 6, 9, 10\}$. If A be the set of all order pairs (x, y) such that x is a factor of y . Then, how many elements does the set A contain? **2012 II**
 (a) 12 (b) 10 (c) 7 (d) 6

- 56.** Consider the following in respect of the sets A and B . **2013 I**

I. $(A \cap B) \subseteq A$ II. $(A \cap B) \subseteq B$ III. $A \subseteq (A \cup B)$

Which of the statement(s) given above is/are correct?

- (a) I and II (b) II and III (c) I and III (d) All of these

- 57.** In a school there are 30 teachers who teach Mathematics or Physics. Of these teachers, 20 teach Mathematics and 15 teach Physics, 5 teach both Mathematics and Physics. The number of teachers teaching only Mathematics is **2013 I**
 (a) 5 (b) 10 (c) 15 (d) 20

- 58.** In a class of 110 students, x students take both Mathematics and Statistics, $x + 20$ students take Mathematics and $x + 30$ students take Statistics. There are no students who take neither Mathematics nor Statistics. What is x equal to? **2013 II**
 (a) 15 (b) 20 (c) 25 (d) 30

- 59.** If A is a non-empty subset of a set E , then what is $E \cup (A \cap \phi) - (A - \phi)$ equal to? **2014 I**
 (a) A (b) Complement of A
 (c) ϕ (d) E

- 60.** If A and B are any two non-empty subsets of a set E , then what is $A \cup (A \cap B)$ equal to? **2014 I**
 (a) $A \cap B$ (b) $A \cup B$ (c) A (d) B

- 61.** Out of 105 students taking an examination English and Mathematics, 80 students pass in English, 75 students pass in Mathematics, 10 students fail in both the subjects. How many students pass in only one subject? **2014 I**
 (a) 26 (b) 30 (c) 35 (d) 45

- 62.** Let A denotes the set of quadrilaterals having two diagonals equal and bisecting each other. Let B denotes the set of quadrilaterals having diagonals bisecting each other at 90° . Then, $A \cap B$ denotes **2015 II**

- (a) the set of parallelograms (b) the set of rhombuses
 (c) the set of squares (d) the set of rectangles

- 63.** Let S be a set of first fourteen natural numbers. The possible number of pairs (a, b) , where $a, b \in S$ and $a \neq b$ such that ab leaves remainder 1 when divided by 15, is **2016 I**
 (a) 3 (b) 5 (c) 6 (d) None of these

- 64.** In a gathering of 100 people, 70 of them can speak Hindi, 60 can speak English and 30 can speak French. Further, 30 of them can speak both Hindi and English, 20 can speak both Hindi and French. If x is the number of people who can speak both English and French, then which one of the following is correct? (Assume that everyone can speak atleast one of the three languages) **2016 I**
 (a) $10 < x \leq 30$ (b) $0 \leq x < 8$ (c) $x = 9$ (d) $x = 8$